

Performance Testing Phase

Project Performance Test

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites (2019)

Project Title	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites (2019)
Domain	Culture & Heritage Analytics / Tourism & Conservation Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Kaggle UNESCO World Heritage Sites Dataset, 1978–2019 (~1,121 sites, 12 fields)
Prepared By	M. Deepika (Team Lead), M. Supraja, K. Srikala, I. Pushpa, U. Bhavana
Mentor	Not yet assigned
GitHub Repository	https://github.com/mankudeepika123-rgb/Heritage-Treasures-An-In-Depth-Analysis-of-UNESCO-World-Heritage-Sites-2019-
Tableau Public Link	https://public.tableau.com/app/profile/manku.deepika5282/viz/HeritageTreasuresAnIn-DepthAnalysisofUNESCOWorldHeritageSites2019/UNESCOSiteOverviewDangerSitesAnalysis
Date	22/08/2026

6.1 Performance Testing Results

Test #	Test Description	Method	Expected Result	Actual Result	Status
PT-01	Amount of Data Rendered	Connect the UNESCO site dataset to Tableau and load any sheet	All rows render without error	~1,121 site rows loaded across 12 fields — no errors	PASS
PT-02	Category Filter Response	Apply Category filter on the distribution chart	Chart updates within 3 seconds	Chart redraw in under 2 seconds	PASS
PT-03	Region / Country Filter Response	Apply Region filter on the World Map	Map updates within 3 seconds	Map redraw in under 2 seconds	PASS
PT-04	Danger Status Accuracy	Verify Danger Status flag matches source data with no null values	No null or error values	Confirmed: Danger Status values plot correctly for all sites	PASS
PT-05	Number of Visualizations	Count all sheets, dashboards, and story points	Charts + dashboards + story as planned	Confirmed: chart sheets, 3 dashboards, 1 story with 3 points, as planned	PASS
PT-06	Cross-Browser Compatibility	Open Tableau Public URL in Chrome, Edge, and Safari	Dashboard loads and filters work in all browsers	Dashboard loaded and all filters responded	PASS

Test #	Test Description	Method	Expected Result	Actual Result	Status
				correctly with no rendering errors	
PT-07	Tooltip Accuracy	Hover over map points, category bars, and trend line points	Correct Site/Country/Category and value shown	Verified: tooltips show correct data on all chart types	PASS
PT-08	Story Navigation	Click arrow buttons to navigate between all story points	Smooth transition; correct dashboard loads at each point	All story points navigate correctly with captions visible	PASS

Test Coverage & Results All eight test cases passed with no defects logged. The two tests most relevant to end-user experience — Category Filter Response (PT-02) and Region/Country Filter Response (PT-03) — both returned well under the 3-second non-functional requirement (NFR-02), confirming the ~1,121-row dataset does not need aggregation to stay responsive on Tableau Public's free hosting tier. Cross-browser testing (PT-06) and Story navigation (PT-08) confirm the dashboard is safe to share as a single public link without a browser-specific caveat.

6.2 Performance Summary

Metric	Value
Total Dataset Rows (Sites)	~1,121
Total Fields	12
Time Range Covered	1978–2019
Number of Filters Used	3 (Category, Region/Country, Danger Status)
Number of Visualizations / Charts	4
Number of Dashboards	3
Number of Story Points	3
Overall Performance Rating	PASS — No performance issues detected

Interpretation The performance metrics above support the scalability requirement defined in Section 2.2: with 3 active filters and 4 core visualizations already running comfortably within the response-time target, there is clear headroom to add further inscription years, additional calculated fields, or an extra dashboard without requiring a paid Tableau Server upgrade. No regressions, broken filters, or rendering failures were observed across any of the eight test cases.